MotionTales: Enhancing Creative Physical Expression in Elementary Education

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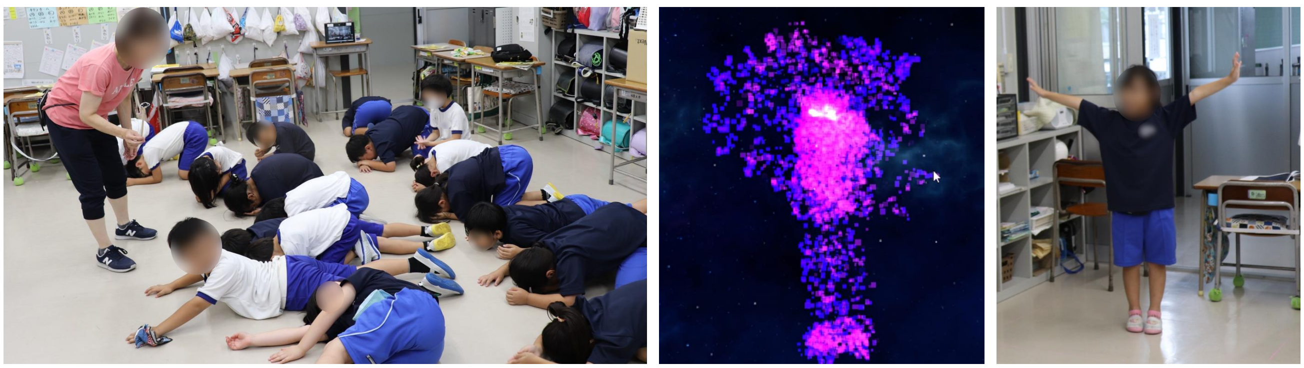

Figure 1: (left) An elementary school storytelling lesson incorporating expressive movement activities. (middle) Visualization of movements based on story-based particle effects. (right) A student expressing the movements of a story character.

Abstract

The expressive movement activities in Japanese elementary education aim to help students express their emotions and thoughts through physical movement. Traditionally, teachers select themes for expressive movements from daily life and guide students, but it is challenging to employ a standardized approach to fostering students' creativity. Consequently, instructional methods often rely on the teacher's subjective judgment. This reliance hinders the development of students' creativity and emotional expression while imposing a significant burden on teachers. We propose MotionTales, an educational tool integrating expressive movement activities with storytelling. This system uses a normal RGB camera and a projector to capture students' gestures and visualize them on a screen using particle effects, and promotes creative improvisational expression. By employing interactive visualizations of diverse story-based movements, we aim to enhance a variety of physical movements to express emotions and thoughts. A user study in a Japanese elementary school has demonstrated that MotionTales increases student engagement in lessons and improves creative expression skills from both physical and linguistic perspectives. Additionally, it provides teachers with a consistent instructional framework, potentially enhancing educational quality. It integrates physical and intellectual activities, offering a new approach to storytelling and dance education in elementary schools.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Child-Computer Interaction, Dance Education, Storytelling, Embodied Interaction, Mixed-Reality

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1 Introduction

Expressive movement activities serve as a primary method for acquiring expressive skills, which is one of the main objectives of elementary education in Japan. The Ministry of Education, Culture, Sports, Science and Technology (MEXT) revised the national curriculum guidelines in 2008, making dance mandatory in elementary schools as of 2011, junior high schools as of 2012, and high schools as of 2013. Expressive movement activities, particularly within the elementary school dance curriculum, emphasize activities designed to enhance physical coordination, creativity, and emotional expression. Specifically, it involves improvisational activities where students express some themes from daily life through body movements.

Unlike predetermined movements in rhythm dance or folk dance, expressive movements allow for a wide variety of motions, making standardized teaching methods challenging. Consequently, teaching methods often rely on the teacher's subjective judgment. Teachers might use visual aids or assign physical tasks, but the choice of materials and teaching methods heavily depends on individual discretion, leading to variability in class quality [Yonezawa 2012]. Moreover, expressing emotions and psychological states through physical movement is abstract and can be difficult to comprehend [Koch et al. 2019], necessitating innovative approaches to elicit diverse expressions from students. This complexity places a significant burden on lesson preparation and execution, leading some teachers to feel inadequate in their instruction [Wakai et al. 2021].

Storytelling, a language-based method, is another approach in elementary education for fostering expression, communication, and creativity. It not only enhances vocabulary, reading comprehension, and writing skills [Wright and Dunsmuir 2019] but also promotes creativity, critical thinking, emotional understanding, and expression and fosters empathy [Monahan et al. 2023]. Traditionally, storytelling is considered an intellectual activity separate from physical activities. Students are expected to sit quietly and listen, with physical activities confined to gymnasium exercises. However, past research indicates the importance of integrating the body and mind to mobilize all cognitive skills and promote active learning [Ionescu and Vasc 2014; Shapiro and Stolz 2019].

Inspired by these ideas, digital tools for multisensory interaction, combining visual, auditory, tactile, and kinesthetic senses, have already emerged in the field of Child-Computer Interaction (CCI). For example, digital tools that use sound to support creative physical expression have been developed, suggesting potential applications in various educational contexts, such as storytelling in kindergartens [Voillot et al. 2024] and improvisational dance in universities [Yamaguchi and Kadone 2017].

We propose a storytelling-based mixed reality educational tool, MotionTales, to explore the effects of a novel approach that combines expressive movement activities and storytelling in elementary education. This system aims to enhance students' creativity and expressive abilities by using diverse story-based themes to express emotions and thoughts through physical movements. This system uses a normal RGB camera and a projector to capture students' gestures and visualize them on a screen with effects, promoting creative improvisational expression. Specifically, students experience and shape stories through their movements, with gestures generating visual effects that align with the narrative. By utilizing interactive motion visualization reflecting various themes based on stories, we aim to enhance students' creativity and expressive abilities through physical movement.

This paper introduces an ongoing interdisciplinary project. First, we outline formative studies to understand the current state and challenges of expressive movement activity classes in elementary schools and to identify the desired features of educational materials. Next, we describe the MotionTales system, which visualizes students' body movements and promotes storytelling activities using a normal RGB camera and projector. Finally, we report the results of a user study conducted in a second-grade public elementary school class in Japan, discussing the findings and future improvement directions.

This work offers the following contributions:

(1) A formative study summarizing the desired content and functions of existing expressive movement activities in elementary education;

(2) Development of a story-based expressive movement support system that can be easily introduced into elementary classes using a normal RGB camera and projector;

(3) Implementation and evaluation of the proposed system in a Japanese elementary school class.

2 Related Work

This is an interdisciplinary project that bridges the fields of education and human-computer interaction. This section first discusses the role of physical expression in education, then introduces interfaces that support creative dance using Extended Reality (XR) technologies, and finally explores interactive educational materials employing digital technologies.

2.1 Effects of Creative Physical Expression

Dance is a form of creative physical expression where dancers convey their imagination and emotions to others [Mahindru et al. 2023]. The close relationship between emotions and bodily movements suggests that applying dance in education and healthcare can offer numerous developmental benefits [Melzer et al. 2019]. Previous studies have demonstrated that creative dance, folk dance, and rhythm dance contribute to the development of social skills [Biber 2016], socio-emotional skills [Panagiotopoulou 2018], and confidence [Odemis and Ilhan 2016] in educational contexts. Additionally, in the healthcare field, dance therapy incorporating imitation and improvisational dance has shown effectiveness as a non-pharmacological intervention, improving depression [Pinniger et al. 2012], reducing anxiety [Adam et al. 2016; Ho et al. 2016], and enhancing cognitive functions [Guerra-Balic et al. 2017].

Thus, dance contributes not only to the improvement of physical motor skills but also to emotional and social development. By fostering body awareness, rhythm, cooperation, and self-expression, dance plays a crucial role in childhood development. Consequently, many countries and regions have integrated dance into their school physical education curricula [Hardman et al. 2014], underscoring its importance in elementary education.

2.2 Interfaces Supporting Creative Dance with XR

Recently, Augmented Reality (AR) and Virtual Reality (VR) technologies have gained attention as tools to inspire creative improvisation in dance. These technologies provide virtual spaces where dancers and choreographers interact with virtual avatars, particles, and objects to enhance spatial awareness and discover new movements and expressions [Raheb et al. 2018; Tsampounaris et al. 2016; Zhou et al. 2023]. Specifically, reflecting dance movements directly in virtual environments creates a clear link between physical movements and virtual effects, enhancing self-efficacy [Lottridge et al. 2022]. Abstract visualizations further promote creative exploration of new movements and expressions [Akbas et al. 2022]. However, these systems are primarily designed for professional or experienced dancers and do not address specific movement tasks or themes, making them less suitable for direct use in elementary education.

Past studies have shown that children with ASD exhibit a strong interest in Mixed Reality (MR) environments [Constantin et al. 2017; Valencia et al. 2019], suggesting that MR can advance dance therapy [Chen et al. 2022; Huang and Lee 2019]. Liu et al. proposed an MR-based dance therapy tool and demonstrated its effectiveness as an educational platform [Liu et al. 2024]. Considering the diverse expressions of students in elementary schools, an MR-based dance support interface could support expressive movement education in these settings.

2.3 Interactive Educational Materials

The integration of digital technologies has made school education more accessible, personalized, and engaging. Interactive technologies such as tablets, smartboards, and VR/AR systems improve learning quality through collaborative learning, virtual field trips, and participatory lessons [Chou et al. 2012]. These technologies support educational practices, engage students, and enhance their motivation to learn. For example, storytelling, a crucial educational practice, has evolved into *"digital storytelling"* through the use of digital technologies. Educational materials combined with AR technology [Tyurina 2023] and tangible user interfaces [Sylla et al. 2015] have also been developed. Given these advantages, we aim to develop versatile interactive educational materials, conducting practical experiments in real classroom environments to evaluate their effectiveness.

3 Formative Studies

To inform the design of our interface, we conducted formative studies consisting of focus groups and an online survey. Our goal was to understand how elementary school teachers conduct expressive movement activity lessons and to prioritize the features to be implemented in the proposed system.

3.1 Focus Groups

We organized focus groups to understand the current state of expressive movement activity lessons and identify potential features for the system. We recruited three focus groups from one public elementary school in Japan, with a total student population of approximately 530. The groups comprised 4, 4, and 5 teachers respectively (5 men, 8 women). The teachers had between 1 and 37 years of teaching experience (M = 12.5, SD = 12.0) and had taught various grades from 1st to 6th in elementary schools. All participants had experience teaching expressive movements and dance. The interviews followed a semi-structured format, where the teachers gathered in a classroom at the school and freely responded to questions from an online interviewer. We began with icebreaker questions (e.g., years of teaching experience, current grade levels taught), followed by discussions on the current content and methods of expressive movement activity lessons, the challenges they face, and what they prioritize in their teaching. We also asked them to share their views on the elements that should be included in the proposed system. From the focus groups, we extracted common practices and issues, leading to several key findings:

- In lower grades, lessons often use easily understandable themes such as mimicking animals (e.g., rabbits, frogs) or using newspapers for expressive play. In higher grades, the focus shifts to group activities based on abstract concepts (e.g., storms, volcanic eruptions).
- Challenges include students feeling embarrassed or freezing up, and difficulty in fostering free expression. Teachers also found evaluation challenging.
- Teachers use onomatopoeia and descriptive words (e.g., fluttering, swaying in the wind) to inspire students' creativity.
- All groups supported using storytelling themes for expressive movement, suggesting it could make lessons more enjoyable, especially for younger students who struggle to sit still. However, they also cautioned against straying too far from the story context.
- Many teachers emphasized the importance of engaging students' interests and encouraging active participation in their lessons. They also highlighted the need to respect students' diversity and to design lessons that bring out individual characteristics.

Teachers frequently used themes like newspapers, jellyfish, and volcanoes, showing a bias in the variety of themes. Moreover, the methods for enhancing and evaluating students' physical expression heavily relied on individual teachers. These findings aligned with our motivation to design a digital tool based on storytelling for expressive movement activities.

3.2 Online Survey

To understand the features most needed by elementary school teachers, we conducted an online survey. Based on the focus group results, we generated seven potential features (Table 1) for the proposed system. The survey asked participants to rate the importance of each feature using a 5-point Likert scale (5: Definitely use the feature; 1: Never use the feature).

We recruited 42 teachers (13 men, 29 women) from seven elementary schools in Japan, with 37 of them having experience teaching expressive movement activities and dance. The teachers had 1-40 years of teaching experience (M = 15.7, SD = 11.8).

3.3 Survey Results and Discussion

Table 1 summarizes the results, including the mean rating and standard deviation for each feature. All features were considered useful, with the lowest average score above 3.5. However, implementing

Table 1: Online survey results: The elements are listed from the highest to the lowest mean score. Five features (F2-F6) are currently implemented in the MotionTales system. F1 was excluded as it is already well-supported by existing systems, and F7 was excluded due to concerns about overwhelming students with too much information.

Feature	Description	Mean (SD)
F1. Background Music	Plays music or ambient sounds matching the story or movements to create an atmosphere.	4.05 (0.69)
F2. Multi-User Mode	Reflects the movements of multiple students in real time for group presentations.	4.02 (0.80)
F3. Single-User Mode	Reflects the movements of a single student in real time for solo presentations.	3.93 (0.77)
F4. Projection Mapping	Projects images onto a screen to visually represent the world of the story.	3.93 (0.70)
F5. Story Text	Displays the story text on the screen in sync with the story progression.	3.88 (0.82)
F6. Visual Feedback	Allows students to see their movements in real-time on the screen to confirm their actions.	3.79 (0.71)
F7. Auditory Feedback	Provides sound effects corresponding to movements, enhancing the sense of accomplishment.	3.57 (0.79)

all proposed features might overwhelm the target users and undermine the core value of the system. Therefore, we excluded a feature already well-supported by existing systems (F1). We focused on features that support students in performing creative physical expressions while experiencing a story displayed on a screen. We narrowed it down to the most robust features based on current technology. A feature scoring below 3.6 was excluded due to significant variation and concerns that excessive information might confuse students (F7). Consequently, we focused on the five features (F2-F6) listed in Table 1.

In summary, these formative studies informed the development of our MotionTales prototype, ensuring it addresses the needs and challenges identified by teachers in the field.

Figure 2: System screen. The top of the screen displays the story text (in Japanese), accompanied by a background that represents the scene described in the text. The center features effects that visualize body movements. The bottom of the screen contains a user interface for necessary operations.

4 MotionTales

4.1 System Design

We designed MotionTales as an application running on a computer using the Unity game engine (Unity 2022.3.35). A projector is used to display the computer's screen onto a classroom screen. The application screen includes the story text (F5), a background representing the scene described in the text (F4), and the visualization of students' movements (F6) (Figure 2).

We envision elementary school teachers using the projected PC screen to create storytelling lessons. Students read aloud the text displayed on the screen and express the content through full-body movements. The screen shows the environment of the story's scene, such as underwater, or in a forest, developed using Unity 3D. We acquired pre-made graphic elements from the Unity Asset Store to enhance the virtual environment. A normal RGB camera captures the students' movements during expressive activities, and these movements are visualized as effects on the screen (detailed in Section 4.2). Students not engaged in the expressive activity can observe both the screen and their peers. This system design, incorporating elements F4-F6, allows the entire class to experience the story together. Additionally, it mitigates the bias in theme variety and the reliance on teacher-dependent methods for enhancing physical expression, as observed in our preliminary study.

4.2 Movement Visualization Design

The primary function of MotionTales is the visual feedback of students' movements during presentations (F6). We use a normal RGB camera to capture the students' movements, which are visualized on the screen using particle effects. The camera is positioned in front of the performing student. We have validated the use of both built-in PC cameras and external cameras, such as an iPhone. For single-camera motion capture, we used MobileNet. The current implementation of MotionTales supports the capture and visualization of movements of 1-10 students simultaneously.

We use particle emitters to visualize students' movements. This particle-based visualization does not overly direct the students' imagination and provides individualized feedback tailored to each student's movements. It allows students to confirm their actions through a quantitative and direct visualization.

4.3 Interaction Modes

Based on our preliminary study results, we created two visualization modes: single-user mode (F3) and multi-user mode (F2). These modes can be switched depending on the scene or objective.

4.3.1 Single-User Mode. Single-user mode is used when a student presents alone. This mode supports solo presentations. It visualizes the movements of a single student through the projector. Depending on the scene, we provide visualization of full-body movements (Figure 3 ⓐ) or upper-body movements (Figure 3 ⓑ).

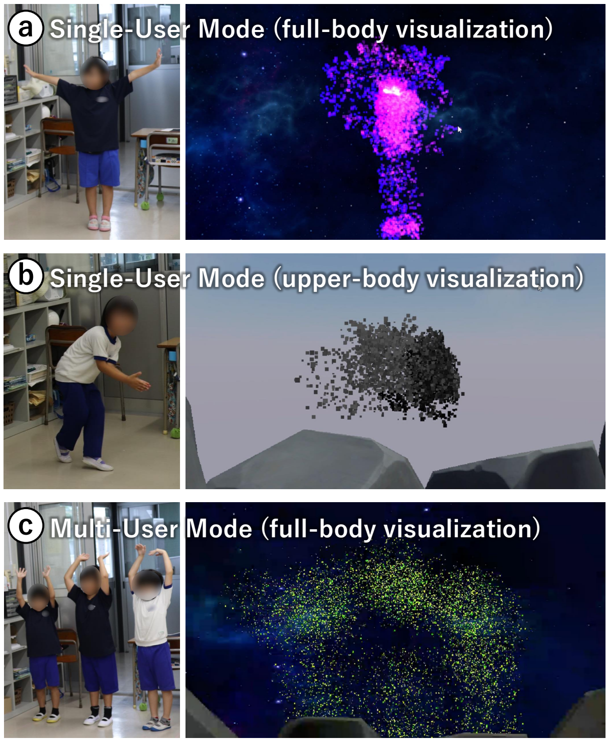

Figure 3: (left) Students engaging in physical expression, and (right) movement visualizations displayed on the screen: ⓐ single-user mode with full-body visualization, ⓑ single-user mode with upper-body visualization, ⓒ multi-user mode with full-body visualization.

4.3.2 Multi-User Mode. Multi-user mode is used when multiple students are presenting simultaneously. This mode supports group work, which is essential in expressive movement activities. We provide visualization of full-body movements (Figure 3 ⓒ).

4.4 Pedagogical Scenario

Based on the results of the formative studies, we developed a pedagogical scenario [Voillot et al. 2024] with elementary school teachers for integrating expressive movement activities and storytelling using the application. Before starting the scenario, students gather in a classroom with enough space for physical expression.

The pedagogical scenario consists of three steps, repeated for each scene of the story:

(1) **Story reading**: All students read aloud the text of the story displayed on the system's screen.

(2) **Expressive performance**: Selected students , either individually or in groups depending on the scene, come to the front of the classroom to perform physical movements that express the story's text. The system visualizes their movements on the screen using particles. The teacher demonstrates the movements if necessary.

(3) **Observation**: The other students observe the presenting students' movements and the visualizations on the screen. While observing, they are free to comment and move their bodies.

5 User Study: in An Elementary School Class

We conducted a user study in a second-grade class at a public elementary school in Japan. The purpose of this study was twofold: 1) to evaluate the technical aspects of the current version of the application, and 2) to assess its qualitative impact in a typical elementary school classroom. The assessment included expressive movement performance, observation, individual interviews with teachers, and surveys with the students.

This study was approved by the ethics committee of the University of Tokyo (#24-20). Participants had to be in good health and willing to take part in an anonymous study. For students, consent was required from the responsible teacher, the school principal, and the parents.

5.1 Paricipants

Twenty-three second-grade students (9 boys, 14 girls) participated in this study. The homeroom teacher, who has 23 years of teaching experience, conducted the lesson. None of the students had been involved in the design of the application or had heard about it before the study.

5.2 Material

We used the story *"Swimmy,"* (by Leo Lionni) found in the second-grade Japanese language textbook, as the material for this study. We extracted parts of the story to create four scenes for a one-hour lesson (as shown in Figure 4). The selection of this story was based on discussions with other teachers during the development phase.

5.3 Procedure

To assess the effectiveness of a typical lesson, the homeroom teacher explained to the students that the session was a regular class, not a special study. We conducted the lesson in the usual classroom. The procedure included implementing the pedagogical scenario (Section 4.4) in each of the four pre-set scenes described in Figure 4, followed by an open-ended feedback survey for the students and semi-structured individual interviews with the teachers.

The homeroom teacher conducted the lesson, with two additional teachers observing and one university student assisting with the system setup. We used a 1.55m×0.95m screen, which was attached to the front blackboard. The lesson was video-recorded, and the recordings were sent to the participating teachers for review before the interviews were conducted. These interviews were audio-recorded. All audio and video recordings were reviewed and annotated by researchers who were absent in the classroom. The entire process took place during a 45-minute lesson in the fifth period of a typical school day.

5.4 Results

5.4.1 Technical Evaluation. From a technical perspective, the system functioned smoothly. Users found the system easy to operate, including screen transitions and controls, without requiring advanced technical knowledge. However, we identified some challenges, such as the lack of realistic visuals and the camera's sensitivity to background influences. These issues will be addressed in future improvements (see Discussion). Additionally, the screen size

Figure 4: Four scenes used in the lesson: ⓐ*"He was scared, lonely, and very sad."*: Using single-user mode, visualized upper body movements with black particles. ⓑ*"Swimmy was happy again."*: Using single-user mode, visualized upper body movements with black particles. ⓒ*"He saw a medusa made of rainbow jelly."*: Using single-user mode, visualized full body movements with pink and purple particles. ⓓ*"A forest of seaweeds growing from sugar-candy rocks."*: Using multi-user mode, visualized full body movements with green particles.

Figure 5: The word cloud illustrates the frequently occurring words from the students' feedback (translated into English). The more frequently a word appears, the larger it is displayed.

was considered too small, with suggestions for larger, classroom-wide projection mapping for a more immersive experience. The addition of auditory feedback, such as sound effects, was also recommended.

5.4.2 Survey Evaluation. We analyzed the students' open-ended feedback survey results using natural language processing (see Figure 5). Words related to specific characters such as *"kelp," "Swimmy,"* and *"jellyfish,"* as well as terms related to movement like *"movement,"* and *"energetic,"* were frequently mentioned. Emotional terms such as *"interesting"* were also common. This indicated that students enjoyed observing and engaging with both their own and others' movements. The frequent mention of *"screen"* also showed that the students were highly interested in the system and paid close attention to the screen. Specific comments included, *"I understood that when I moved, (Swimmy on) the screen did the same thing. I felt like I could become Swimmy."* Another student noted, *'When XX (a student's name) spread her arms wide, it looked like wings were growing on the screen."* These comments suggested that the system's visualization supported the imagination of both the performing students and their observers.

5.4.3 Qualitative Analysis of Video Recordings. All annotators independently noticed that students actively participated in interactive storytelling and enjoyed their movements, as evidenced by their smiles and laughter. Students presenting in front of the class exhibited moments of hesitation and contemplation while visualizing their movements on the screen but continued to perform creative dance movements (e.g., twirling their arms, rising from a crouched position, swaying their entire bodies) by referencing comments from observing students and imitating the teacher's actions. This behavior indicates the achievement of learning objectives for expressive movement activities in elementary schools. Students observing the presentations imitated the movements, provided comments, and applauded, indicating active engagement across the class. Comments included positive remarks like *"Well done!"* and *"It's like a jellyfish!"* as well as negative ones like *"Not similar."* and *"Not kelp."* (see Discussion). Some students used creative metaphors such as *"Like a donut!"* and *"Like a banana!"* Overall, many students observed with smiles, creating a supportive atmosphere for the dancers.

In single-user mode (F3), presenters eagerly stepped forward with smiles, indicating their enthusiasm. Other students showed keen interest in the presenter's movements, calling out names and cheering, displaying high expectations for the selected student. In multi-user mode (F2), many students eagerly raised their hands to be chosen, suggesting a lower threshold for involvement compared to the single-user mode. In both interaction modes, students actively raised their hands to participate. As the lesson progressed, students began to understand how to perform theme-based dances. Initially captivated by the screen, they gradually focused more on creatively executing their dance movements.

5.4.4 Teacher Interviews. We conducted interviews with three teachers: the homeroom teacher who led the lesson (Teacher 1) and two teachers who observed the lesson (Teacher 2 and Teacher 3). All teachers agreed that the system was educationally beneficial, especially for younger students, as it helped them understand concrete objects. This benefit was evident in the students' high engagement and enjoyment during the lessons compared to regular classes. [*"The students were very interested in the system and found joy and surprise in seeing their movements visualized."* - Teacher 2] Teachers also appreciated the ability to view their students from a different perspective. [*"Students who usually struggle to verbalize their* *thoughts were actively participating this time. They seemed to enjoy thinking through physical expression, even if they found verbalization difficult."* - Teacher 1]

Teachers emphasized the importance of realistic experiences facilitated by the screen's visuals in bridging the gap between the story and the students. They noted that virtual environments helped understand the narrative but were not widely used in elementary schools. [*"Some students have never been to the sea, making it difficult for them to immerse themselves in the story by just reading. Visuals help support their imagination."* - Teacher 3]

Additionally, teachers pointed out that real-time visualization of movements on the screen was useful for improvisational choreography in creative dance. This constant visual feedback encouraged the continuous creation of new movements. They also highlighted the need for integrating the pedagogical scenario, noting that while the system engaged students, its connection to Japanese language lessons remained a challenge. [*"I think expressing texts through movement and eventually verbalizing them could enhance learning effects more."* - Teacher 3]

6 Discussion

This project aims to design and implement a storytelling application for expressive movement activities in elementary schools. We developed an interactive system based on physical movement through formative studies with elementary school teachers and evaluated it in real classroom settings. This section discusses the lessons learned from the formative studies and user study evaluations.

6.1 System Effectiveness and Potential Deployment

The user study in an elementary school class confirmed that the system functions effectively in classroom settings. It was rated as easy to use, requiring no special technical knowledge, indicating its potential for use in other grades and schools. Currently, each story requires a new application design. Future improvements should include making the application more versatile, allowing teachers to easily add stories from language textbooks.

Overall, the user study indicated that the MotionTales system could encourage students and teachers to try different methods. As many teachers noted, lessons using MotionTales differ significantly from traditional storytelling, which typically involves reading and understanding texts while seated. Teachers observed that integrating expressive movement activities into storytelling lessons helped students who struggled with verbalization to express and understand characters' emotions through physical actions. Moreover, it increased students' motivation and enjoyment of learning.

Typically, storytelling lessons in elementary schools take about 10 to 20 hours to complete a single story. In this experiment, we used the system after completing the story *"Swimmy."* Teachers suggested various timings for using the system: [*"Using it at the very beginning can quickly enhance students' imagination about the story."* - Teacher 3] [*"During different scenes of the story, where words alone may not fully convey the idea, visual and physical activities can support their imagination."* - Teacher 1] Future evaluations should assess the system's effectiveness at different stages over a longer period and explore adaptation for a wider range of ages.

From a usability perspective, teachers noted that MotionTales could be used throughout the year in various educational contexts. They suggested that visualizing physical movements could be useful not only for storytelling but also in other subjects: [*"It could be used in music classes to think about movements while singing."* - Teacher 3] [*"It could also be applied in gym classes for expressive movement activities."* - Teacher 2] At the developmental stage, elementary school students express themselves not only verbally but also nonverbally. The visualization of movements through MotionTales could serve as a central teaching material to develop activities such as acting out stories together, choreographing dances to music, collaborating to create movements, and mimicking or commenting on friends' actions.

6.2 Improvements and Future Challenges

As previously mentioned, the system has several areas for improvement. First, the simple particle system used in this study did not realistically replicate movements such as those of kelp and jellyfish, which limited the expansion of students' imagination. This limitation may have led to the negative comments observed among the students watching the presentations. Future designs should focus on creating a system that can visualize students' movements more realistically. However, during the preliminary survey, several teachers emphasized the importance of ensuring that visual aids do not hinder students' ability to read and expand their imagination from the text. Over-reliance on visual aids might restrict their imagination, potentially causing the opposite of the desired effect. Therefore, we should consider the context in which visual aids are used, focusing on specific parts to be visualized. Additionally, we did not incorporate auditory feedback in our system, although several studies have demonstrated the positive effects of sound on creativity in bodily expressions [Voillot et al. 2024; Yamaguchi and Kadone 2017], which should be considered in future research.

Furthermore, we did not conduct a quantitative evaluation of expressive movements in this study. In the future, we aim to quantitatively measure the improvement in creativity in expressive movements by utilizing motion estimation technology and other quantitative methods.

7 Conclusion

In this study, we introduced and evaluated MotionTales, a storytelling-based MR educational tool designed to enhance elementary students' creative physical expression abilities. MotionTales promotes creative activities by projecting narratives onto a screen, allowing students to embody characters through physical expression. The user study confirmed that the system operates smoothly and is easy to use, though improvements in visual effects, camera accuracy, and screen size are needed. We observed high levels of student interest and engagement, demonstrating that the visualization of movements stimulated their creativity and imagination. Teachers found the system particularly beneficial for younger students and those struggling with verbal expression, noting that real-time movement visualization enhanced story comprehension and participation. Future developments should focus on integrating various stories seamlessly and exploring applications in other educational contexts. MotionTales integrates physical and intellectual activities, offering a novel approach to storytelling and dance education in elementary schools. Finally, while we designed this application to meet specific goals of the Japanese elementary curriculum, we believe it can be easily adapted and applied to other educational contexts, including those involving students with disabilities and in different countries.

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